

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Dec 2023

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

QP No. :UD160-1

Duration: 3 hours

Max. Marks:100

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. Find the value of $\lim_{x \rightarrow \infty} \sinh^{-1} x - \log x$.
2. Express $\cos^2 \theta \sin \theta$ in ascending powers of θ as far as the term involving θ^6 .
3. Check the matrix $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ is orthogonal. $AA^T = A^T A = I$
4. Using system of linear equations, balance the chemical equation $H + O_2 \rightarrow H_2O$.
5. Solve $x^4 + 4x^3 + 5x^2 + 2x - 2 = 0$ given $(-1 + i)$ is a root.
6. Solve $6x^3 - 11x^2 + 6x - 1 = 0$ given that roots are in H.P.
7. Evaluate $\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta$.
8. Write the connection between pi function and Gamma function.

$$\pi \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

9. When x is small find the value of $\sqrt{x^2 + 4} - \sqrt{x^2 + 1}$. (up to power 4 only.)

10. Solve $x^3 - 4x^2 + 5x - 2 = 0$ given that it has double root.

$$1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

PART- B

Answer all the questions

4 x 15 = 60 Marks

11. a) Expand $\cos^4 \theta \sin^3 \theta$ in terms of multiples of θ . (8)
 b) If $\sin(A + iB) = \cos \theta + i \sin \theta$, prove $\cos^2 A = \sinh^2 B$; $\cos^2 A = \pm \sin \theta$. (7)

(OR)

12. a) i) Find the real and imaginary parts of $i^{\log(1+i)}$.
 ii) Find the value of i^i . (8)
 b) Express $\cosh^6 \theta$ in terms of hyperbolic cosines of multiples. (7)

13. a) Investigate for what values of a, b the simultaneous equations $x + y + 2z = 2$; $2x - y + 3z = 2$; $5x - y + az = b$ will have Rouché's theorem cases. (8)

- b) Solve by singular value decomposition method: $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$ (7)

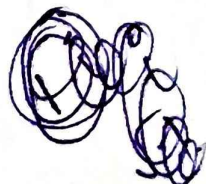
(OR)

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14. a) Reduce the quadratic form to its normal form by an orthogonal reduction. $q = 3x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_2x_3$. (8)

- b) Find the characteristic equation of two matrices and interpret the result. $A = \begin{pmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$. (7)

15. a) Solve $x^4 - 2x^3 + 4x^2 + 6x - 21 = 0$; given that two of its roots are equal in magnitude but opposite in sign. (8)



- b) Using Routh criterion determine the stability of the system generated by characteristic equation $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$. Comment on the location of the roots. (7)

(OR)

16. a) Increase by 2 the root $x^4 - x^3 - 10x^2 + 4x + 24 = 0$ and hence solve them. (8)
- b) Sum to infinity the series $\frac{1^4}{1!} + \frac{2^4}{2!} + \frac{3^4}{3!} + \dots$. (7)

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17. a) Calculate the value of $\int_0^\infty \frac{t^2 dt}{1+t^4}$. (8)
- b) Explain by illustration: a double integral the mass of a circular plate where the density at any point varies as the square of its distance from a fixed diameter. (7)

(OR)

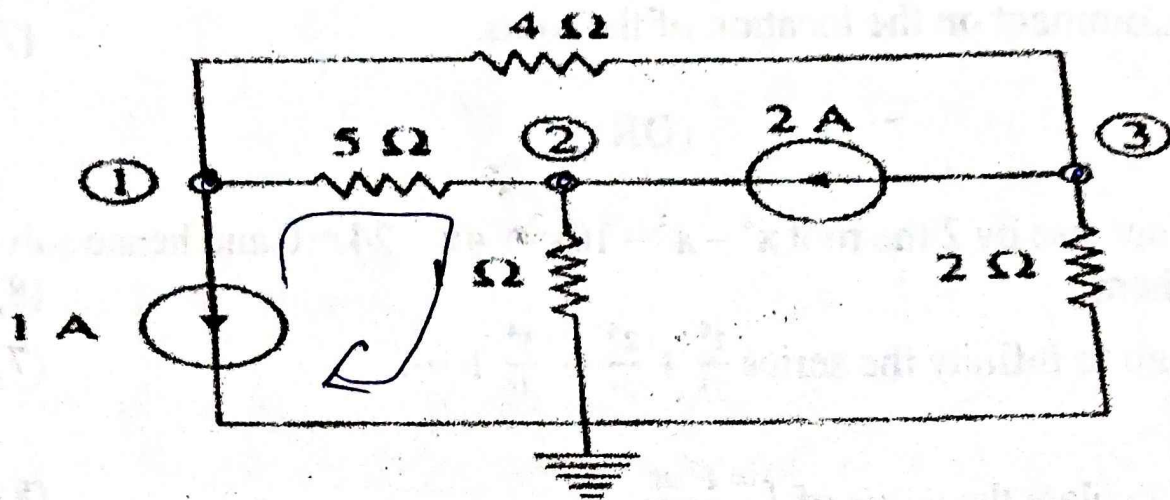
18. a) Evaluate $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} xy \, dy \, dx$ by change of order of integration. (8)
- b) Evaluate $\iiint x^2 yz \, dx \, dy \, dz$ over the tetrahedron bounded by the planes $x = y = z = 0$; $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (7)

PART-C

Answer the following

1 x 20 = 20 Marks

19. a) Calculate the node voltages in the circuit given below by solving the resulting simultaneous equations by matrix inversion method. (10)



b) Find the sum to infinity of the series $\frac{7}{72} + \frac{7.28}{72.96} + \frac{7.28.49}{72.96.120} + \dots$ (10)

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